

Kathmandu Air Quality Intelligence

*Statistical Analysis, Time-Series Modeling, Extreme-Event Analysis, and
24-Hour PM2.5 Forecasting*

Detailed Project Documentation

[Live interactive dashboard](#)

[Link to Google Collab](#)

Study period: September 2022 - April 2025
23,352 hourly observations | 973 daily observations

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1. Executive Summary

Kathmandu Air Quality Intelligence is an end-to-end research and forecasting project that combines hourly pollution observations with meteorological data to characterize temporal structure, quantify statistical associations, identify extreme pollution episodes, and evaluate 24-hour-ahead PM2.5 forecasts.

The integrated dataset contains 23,352 continuous hourly observations from 1 September 2022 through 30 April 2025. The analysis was intentionally structured as a reproducible ten-notebook workflow, with later phases consuming saved outputs from earlier phases rather than relying on notebook kernel state.

- Winter was the highest-pollution season; monsoon was the lowest. Seasonal differences were large and statistically supported.
- PM2.5 was strongly persistent: lag-1 hourly correlation was about 0.965, and lag-24 correlation rebounded to about 0.877, revealing a pronounced daily cycle.
- The top-10% PM2.5 threshold was 66.1. It produced 303 episodes; 254 lasted at least two hours, and the longest lasted 65 hours.
- Meteorological relationships changed materially after temporal and multivariable adjustment, demonstrating strong confounding by season and pollution persistence.
- A leakage-safe Ridge model achieved test MAE 11.31 versus 12.09 for 24-hour persistence, a 6.4% MAE improvement in a substantially higher-pollution test regime.
- Ridge added the most value when pollution changed materially over 24 hours, but it still underpredicted the highest observed concentrations.

2. Research Scope and Questions

The project was organized around five research questions:

Question	Research question	Operationalization
RQ1	What temporal patterns exist in Kathmandu air pollution?	Hourly, weekday/weekend, monthly, seasonal, annual and long-term patterns.
RQ2	Which meteorological conditions are associated with pollution?	Temperature, humidity, rainfall, wind, pressure and dew point versus major pollutants.
RQ3	How persistent is PM2.5?	Lag correlations, ACF/PACF, stationarity, rolling persistence and multi-seasonal structure.
RQ4	What conditions characterize extreme pollution events?	Top-10%/top-5% PM2.5 episodes, AQI episodes, duration, timing, seasonality and weather.
RQ5	Can PM2.5 be forecast 24 hours ahead?	Leakage-safe forecasting against a strong same-hour-previous-day persistence baseline.

The project distinguishes descriptive findings from inferential conclusions and from predictive performance. Observational meteorological associations are not presented as causal effects.

3. Data and Project Architecture

3.1 Integrated variables

Domain	Variables
Pollution	PM2.5, PM10, NO2, SO2, CO
Meteorology	Temperature, rainfall, wind direction, wind speed, relative humidity, surface pressure, dew point
Derived	Calendar/cyclic features, seasons, rain flags, wind-sector encoding, PM2.5/PM10 ratio,

	percentile flags, particulate AQI
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3.2 Data coverage

Property	Value
Hourly observations	23,352
Daily observations	973
Start	2022-09-01 00:00
End	2025-04-30 23:00
Inferred hourly frequency	1 hour
Duplicate timestamps	0
Missing timestamps	0
Missing source values after integration	0

3.3 Repository design

Raw files are preserved separately from interim and processed data. Each analysis phase writes reusable tables or datasets to disk, allowing any later notebook to be restarted independently. Models and dashboard artifacts are also separated from notebooks.

Directory	Purpose
data/raw	Original pollution and weather exports
data/interim	Merged hourly dataset
data/processed	Clean, feature-engineered and daily datasets
data/processed/ml	Frozen forecasting splits
notebooks	Ten analysis phases plus optional Colab hub
reports/tables	Saved numerical outputs by phase
reports/figures	Publication/dashboard figures where required
models	Serialized forecasting model
dashboard	Streamlit application

4. Data Quality and Validation

Phase 2 treated statistical outliers and invalid measurements as different concepts. High pollution values were not removed merely for exceeding IQR thresholds because extreme pollution is a core research target.

Check	Result
Missing values	0
Infinite values	0
Duplicate timestamps	0
Negative pollutant rows	0
Invalid humidity/wind/rain/pressure rows	0
Rows removed	0

Pollutant distributions were strongly right-skewed. PM2.5 had mean 35.05, median 29.0, standard deviation 23.25 and maximum 170.4. Rainfall was zero-heavy, with roughly 78% zero observations, motivating both binary rain-status and amount-based analysis.

A key provenance limitation is that pollutant unit symbols were mangled in the original export headers. The project therefore does not silently assert exact pollutant units in final documentation until the original data-source metadata is verified.

5. Feature Engineering and Particulate AQI

Phase 3 created four major classes of derived information:

- Calendar and season variables: year, month, quarter, weekday, hour, weekend status, time of day and climatological season.
- Cyclical encoding: sine/cosine representations for hour, day of week, month, day of year and wind direction.
- Pollution/weather context: rain status, temperature-dew-point spread, PM2.5/PM10 ratio and extreme percentile flags.
- Daily aggregation: pollutant means, weather summaries, rainfall total and circular mean wind direction.

5.1 Particulate AQI

The project calculates a reproducible particulate AQI from daily PM2.5 and PM10. Current U.S. EPA PM2.5 breakpoints were applied consistently across the complete 2022-2025 study period. Gas AQI components were deliberately excluded because the source export did not provide sufficiently trustworthy unit/averaging-period metadata for those gases.

AQI summary	Result
Good days	21
Moderate days	555
Unhealthy for Sensitive Groups	252
Unhealthy	145
Days particulate AQI > 100	397
Days particulate AQI > 150	145
Dominant particulate pollutant	PM2.5 on all 973 study days

6. Exploratory Findings

6.1 Diurnal pattern

Hourly PM2.5 showed a pronounced daily cycle. Mean concentration reached its minimum around midday and rose sharply through the evening, with the highest mean around 22:00 in the supplied timestamp scale. Because the processed timestamps are timezone-naive, the original source timezone should be verified before interpreting these hours as definitive Kathmandu local time.

6.2 Seasonal and monthly pattern

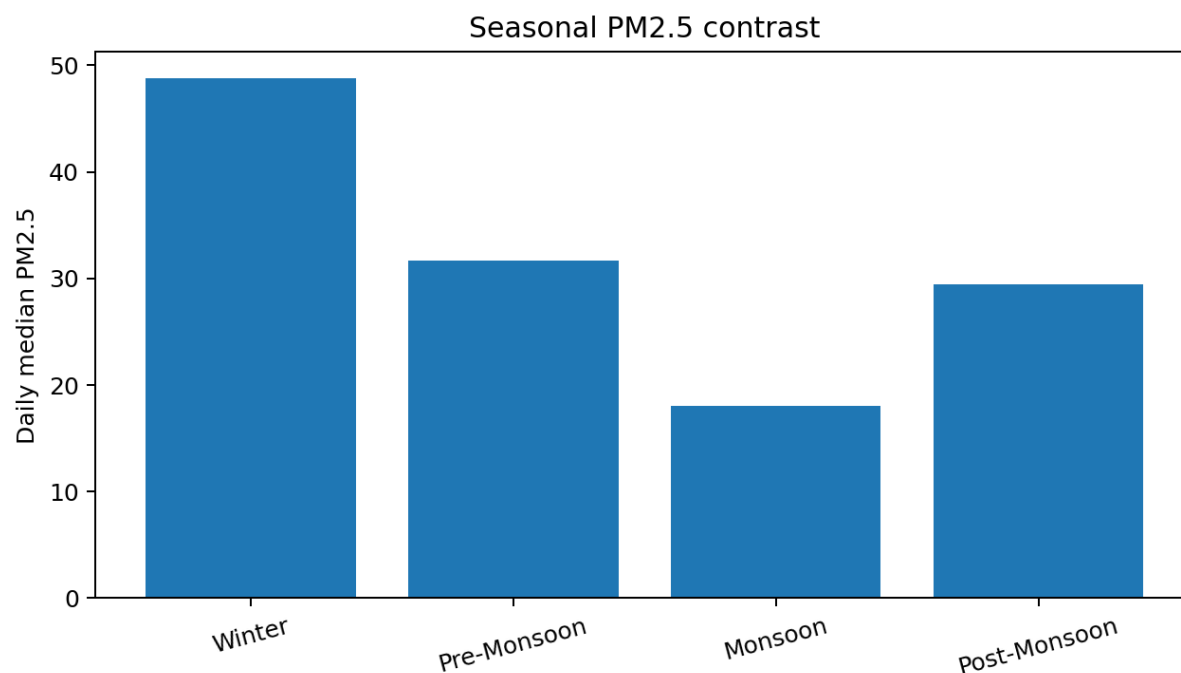


Figure 1. Daily median PM2.5 by climatological season.

Winter daily median PM2.5 was 48.82, compared with 31.68 in pre-monsoon, 18.06 in monsoon and 29.46 in post-monsoon. January had the highest monthly mean (about 55.35), while July had the lowest (about 14.95).

6.3 Pollutant and weather associations

PM2.5 and PM10 were very strongly related (Pearson about 0.949; Spearman about 0.979). Daily meteorological relationships were strongest for temperature, dew point, rainfall and humidity, but Phase 6 showed that several raw associations were substantially confounded by seasonal and temporal structure.

7. Time-Series Structure and Persistence

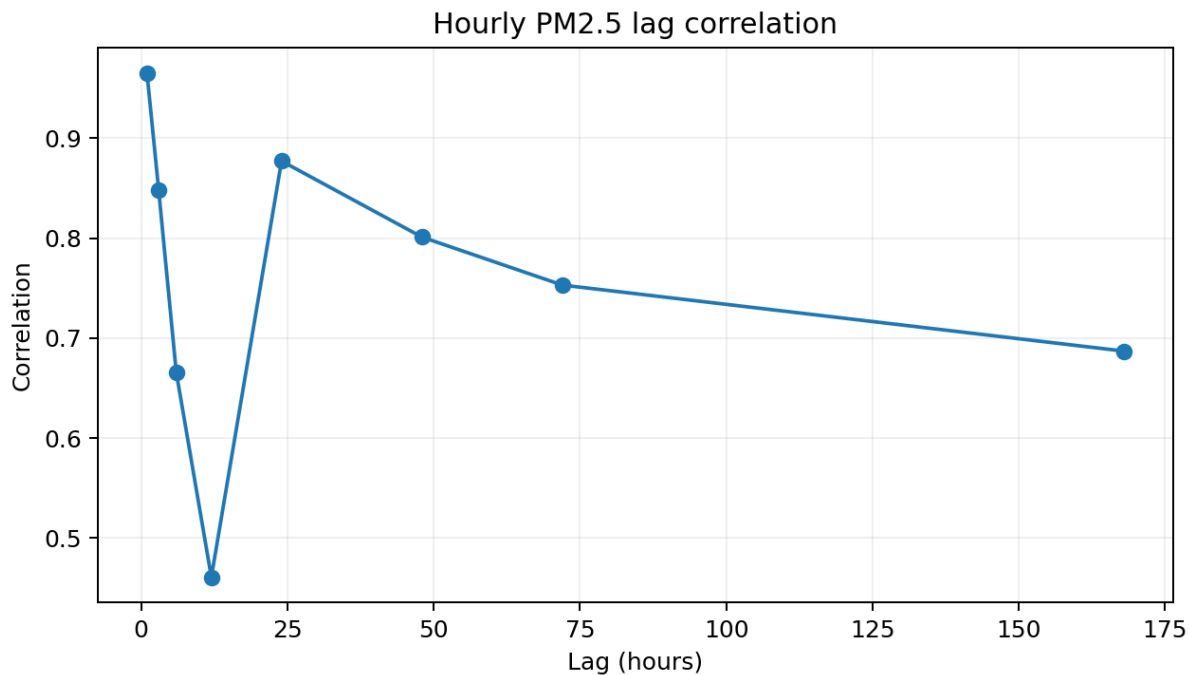


Figure 2. Selected hourly PM2.5 lag correlations.

The hourly series was extremely persistent at short lags. Correlation declined from 0.965 at one hour to 0.461 at 12 hours, then rose again to 0.877 at 24 hours. This rebound is consistent with the pronounced diurnal cycle. Correlation remained 0.687 at one week.

Lag	Hourly correlation
1	0.965
3	0.848
6	0.666
12	0.461
24	0.877
48	0.801
72	0.753
168	0.687

Daily PM2.5 was also highly persistent: lag-1 correlation was 0.919, lag-7 was 0.656, lag-30 was 0.557 and lag-60 was 0.330. ADF/KPSS testing supported non-stationarity in the raw daily series and stationarity after first differencing. However, Ljung-Box testing showed that significant serial dependence remained even after differencing.

Multiple-seasonal decomposition of the hourly series identified a strong 24-hour component and a smaller weekly component. A 365-day STL decomposition was not used for substantive inference because the study spans fewer than three complete annual cycles.

8. Statistical and Meteorological Analysis

8.1 Seasonal inference

Statistic	Result
Kruskal-Wallis H	420.076
p-value	< 0.0001
Epsilon-squared	0.430 (large)
Only non-significant seasonal pair	Pre-Monsoon vs Post-Monsoon
Weekday vs weekend	No meaningful difference
Day-of-week test	p about 0.988; effect effectively zero

The inferential results confirm the descriptive hierarchy: season is a major source of PM_{2.5} variation, while day-of-week is not. Season was also strongly associated with particulate AQI category (Cramer's V about 0.349).

8.2 Confounding in weather relationships

Several raw meteorological correlations changed after controlling for month, hour, recent PM_{2.5} and other weather variables using HAC/Newey-West standard errors. The strong negative raw temperature relationship largely disappeared after adjustment. Wind speed reversed from a positive raw correlation to a negative adjusted coefficient, demonstrating why raw correlations should not be interpreted in isolation. Rain remained negatively associated in both hourly and daily adjusted models, while pressure retained a positive adjusted association.

Previous PM_{2.5} was one of the strongest statistical predictors, consistent with the persistence analysis. Substantial residual autocorrelation remained, supporting the use of HAC inference and motivating time-aware forecasting models.

9. Extreme Pollution Events

Extreme events were defined retrospectively from the study-wide PM_{2.5} distribution. The top-10% threshold was 66.1 and the top-5% threshold was 83.6. Consecutive threshold exceedances were grouped into episodes rather than treated as independent extreme hours.

Extreme-event metric	Top 10%	Top 5%
Extreme hours	2,354	1,177
Episodes	303	188
Sustained episodes (≥ 2 h)	254	162
Median duration	6 h	5 h
Mean duration	7.77 h	6.26 h
Maximum duration	65 h	18 h
Median episode peak	81.3	102.35

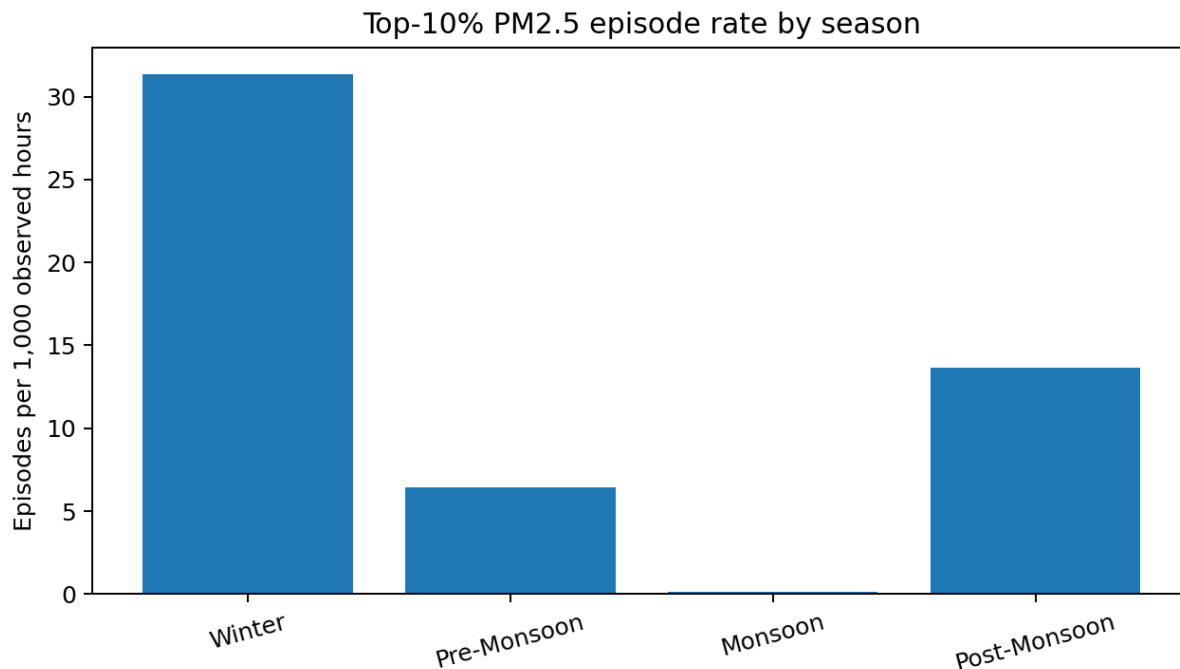


Figure 3. Top-10% PM2.5 episode-start rate normalized by observed hours.

Winter recorded 31.37 top-10% episode starts per 1,000 observed hours, compared with 13.66 in post-monsoon, 6.46 in pre-monsoon and only 0.15 in monsoon. Episodes commonly began in the evening. Southerly wind observations had the highest conditional extreme-hour percentage (about 13.9%), while rainy hours were extreme only about 3.2% of the time versus about 12.0% of dry hours.

Calculated particulate AQI > 100 occurred on 397 days grouped into 56 consecutive-day episodes; AQI > 150 occurred on 145 days grouped into 24 episodes. The longest such periods lasted 112 and 66 days, respectively.

10. Forecasting Dataset and Leakage Control

The machine-learning objective was deliberately formulated as a true forecast: use information available at forecast origin t to predict PM2.5 at $t+24$ hours. Actual observed weather at $t+24$ was excluded because it would not be known at forecast time unless an external weather forecast were supplied.

Feature class	Examples
Current state	PM2.5, PM10, NO2, SO2, CO and current meteorology
PM2.5 lags	1, 3, 6, 12, 24, 48, 72, 168 hours
Rolling history	6, 12, 24, 72 and 168-hour means/standard deviations
Momentum	1h, 6h and 24h PM2.5 change
Co-pollutant history	1h and 24h lags
Weather history	Previous-24h weather means, rain total and rain hours
Known future calendar	Target hour/day-of-week/day-of-year cyclic encodings

After feature construction, 23,160 forecasting rows and 55 predictors remained. The chronological split used 70% training, 15% validation and 15% test with 24-hour purge gaps between adjacent sets. Five-fold expanding-window TimeSeriesSplit with a 24-hour gap was used for model tuning. Scalers were fitted inside model pipelines only on training folds.

The held-out test period exhibited a major distribution shift: mean target PM2.5 was 59.25 compared with 31.30 in training and 28.91 in validation. This makes the final test a genuine stress test under a higher-pollution regime.

11. Machine-Learning Model Comparison

Three model families were compared with the 24-hour persistence baseline: Ridge regression, Random Forest and Histogram Gradient Boosting. Model selection used MAE as the primary metric, supported by RMSE, R-squared and skill relative to persistence.

Model	CV MAE	Validation MAE
Persistence (24h)	7.339	7.212
Ridge	8.587	7.224
Random Forest	7.575	7.903
Histogram Gradient Boosting	8.274	8.235

Persistence was the best average MAE benchmark during cross-validation and validation. Ridge was the best machine-learning candidate on validation and was therefore frozen before the test set was inspected. Its selected regularization parameter was $\alpha = 100$, indicating that strong shrinkage was beneficial in the highly correlated 55-feature design.

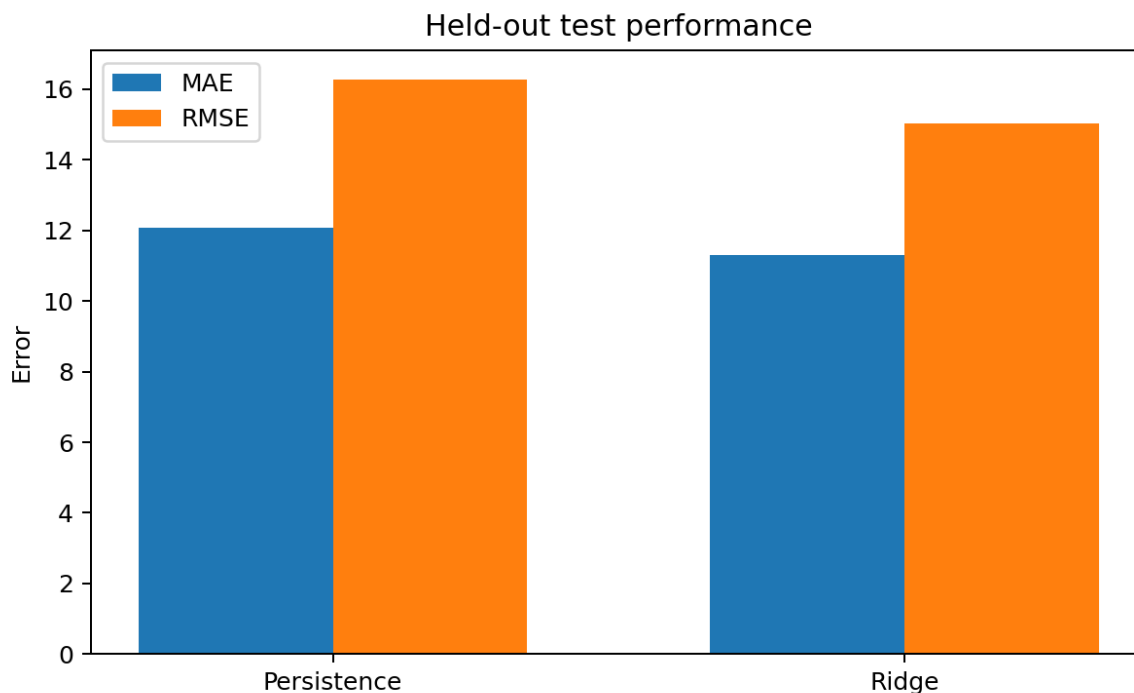


Figure 4. Final held-out test performance of persistence and the frozen Ridge model.

Model	Test MAE	Test RMSE	Test R-squared
Persistence	12.085	16.284	0.738
Ridge	11.314	15.028	0.777

Ridge reduced test MAE by approximately 6.4% and RMSE by approximately 7.7% relative to persistence. Importantly, the model was selected without using test data, so this gain is genuine held-out evidence rather than test-driven tuning.

12. Explainability and Forecast Error Analysis

Phase 10 treated explainability as post-hoc diagnostics. No feature selection, hyperparameter changes or retraining were performed after inspecting the held-out test results.

12.1 What drives the Ridge forecast?

The strongest standardized coefficients and permutation scores were dominated by recent pollution dynamics: current PM2.5, one-hour and 24-hour PM2.5 history, PM10 history, short-term PM2.5 change and medium-term rolling summaries. The most important individual permutation feature was the 6-hour PM2.5 change, followed by current PM2.5 and lag-1 PM2.5.

Individual coefficient signs are not interpreted causally because many predictors are highly correlated. Ridge distributes predictive information across overlapping lag, rolling and co-pollutant features.

12.2 When does Ridge beat persistence?

Absolute 24h PM2.5 change	Ridge MAE	Persistence MAE	Ridge MAE skill
0-5	5.11	2.37	-116.0%
5-10	6.92	7.42	+6.7%
10-20	12.38	14.43	+14.2%
20-40	22.42	26.77	+16.3%
40+	43.53	51.52	+15.5%

This result explains the overall forecasting story. Persistence is ideal when tomorrow resembles today; Ridge becomes useful when pollution conditions change. Ridge also performed better than persistence in elevated, high, very-high and extreme concentration regimes, while persistence was better in low/typical conditions.

12.3 Main limitation: regression toward the mean

Overall Ridge bias was small, but conditional bias changed with concentration. Low concentrations were overpredicted, while very high and extreme concentrations were underpredicted. Among development-defined high-pollution observations, Ridge had lower MAE than persistence but greater negative bias. High-pollution precision improved slightly, recall decreased slightly, and both approaches had approximately the same F1 score.

Forecast residuals retained strong serial autocorrelation through multiple lags. This indicates that predictable temporal structure remains and motivates future sequence-aware or residual-correction models.

13. Interactive Dashboard

The complete project is exposed through a Streamlit application: <https://air-quality-intelligence-kathmandu.streamlit.app/>

The dashboard contains seven sections:

- Overview - headline metrics, daily trend, seasonality and particulate AQI.
- Temporal Patterns - diurnal/monthly patterns, season-hour heatmap, persistence and stationarity.
- Meteorology & Statistics - interactive pollutant/weather plots, correlations and adjusted models.
- Extreme Events - thresholds, episode durations, normalized seasonal rates, wind/rain and AQI episodes.
- Forecasting - Ridge versus persistence, selectable test-period window and subgroup performance.
- Explainability & Errors - coefficients, permutation importance, calibration, residuals and worst misses.
- Methods & Data Quality - reproducibility notes, limitations and variable summaries.

14. Reproducibility and Colab Workflow

14.1 Notebook execution order

The ten notebooks should be executed in numerical order for a full rebuild. However, each phase reads saved files rather than relying on variables from a previous kernel, so later notebooks remain independently restartable once their prerequisite files exist.

14.2 Recommended Colab publication pattern

Google Colab uses individual Jupyter notebook files. For an experience similar to a multi-sheet workbook, the repository should expose one hub notebook named `00_START_HERE.ipynb`. The GitHub README links only to this hub; the hub then links the ten phase notebooks in order. Each phase notebook can include Home, Previous and Next navigation links plus a setup cell that clones the repository and installs dependencies when executed in Colab.

1. Push all notebooks to the public GitHub repository with outputs retained where practical for immediate viewing.
2. Add notebooks/00_START_HERE.ipynb and set the GitHub username in its configuration cell.
3. Run the provided `prepare_colab_notebooks.py` script once to add navigation and Colab setup cells to phases 1-10.
4. Commit the modified notebooks and use a single Open in Colab link to `00_START_HERE.ipynb` from the README.
5. Keep a `requirements.txt` at repository root so every Colab runtime can reproduce dependencies.

Colab runtimes are ephemeral; therefore every independently shared notebook should install its dependencies and obtain required repository files programmatically rather than assuming a prior notebook was executed in the same runtime.

15. Limitations

- Pollutant unit symbols were corrupted in the exported headers; exact unit claims require verification against original provider metadata.
- The processed timestamps are timezone-naive. Hour-of-day findings depend on the supplied clock values and should be tied to the original source timezone before formal publication.
- The particulate AQI is a research calculation based on PM2.5 and PM10, not a complete official multi-pollutant Kathmandu AQI.
- The dataset spans fewer than three full annual cycles, limiting stable annual-seasonality decomposition and long-term trend inference.
- Meteorological findings are observational and remain vulnerable to unmeasured confounding; they should not be described as causal effects.
- Future observed weather was intentionally excluded from the 24-hour forecast. A production system would ideally incorporate real weather forecasts available at forecast origin.
- The final Ridge model still underpredicts some rapid high-pollution surges and leaves substantial autocorrelation in forecast residuals.
- The test period is a higher-pollution regime than development data. This is useful for stress testing but means a single held-out period cannot establish universal superiority over persistence.

16. Future Work

- Verify and document original data provider, pollutant units and timestamp timezone.
- Incorporate actual forecast meteorology rather than observed future weather.
- Evaluate residual autoregressive correction, dynamic regression and sequence-aware models using a new untouched evaluation period.

- Add probabilistic or quantile forecasts to communicate uncertainty and improve high-pollution risk assessment.
- Investigate regime-specific models for stable versus rapidly changing pollution conditions.
- Add event-onset prediction as a separate classification task for new extreme episodes.
- Expand the temporal record and, where possible, validate findings against independent ground-monitoring observations.
- Package reusable preprocessing/modeling logic from notebooks into src/ modules with automated tests.

17. Notebook Map

Notebook	Role	Primary output
00_START_HERE.ipynb	Colab/GitHub navigation hub	Links all analysis phases; no scientific computation.
01_data_loading.ipynb	Data integration	Load, rename, align, merge and validate timestamps.
02_data_quality_cleaning.ipynb	Quality assessment	Physical checks, distributions, outliers and clean dataset.
03_feature_engineering.ipynb	Feature engineering	Calendar/cyclic features, daily aggregation and particulate AQL.
04_exploratory_data_analysis.ipynb	EDA	Distributions, seasonality, diurnal patterns, correlations and event exploration.
05_time_series_analysis.ipynb	Time-series diagnostics	Lag correlations, ACF/PACF, stationarity, persistence and decomposition.
06_statistical_meteorological_analysis.ipynb	Inference	Nonparametric tests, effect sizes, FDR/Holm correction and HAC regression.
07_extreme_pollution_event_analysis.ipynb	Extreme events	Episode construction, severity, timing, weather and AQL sequences.
08_ml_feature_engineering.ipynb	Forecast preparation	Leakage-safe target/features, purged chronological splits and baselines.
09_ml_model_training.ipynb	Forecast modeling	Cross-validation, model tuning, frozen test evaluation and model serialization.
10_explainability_error_analysis.ipynb	Post-hoc diagnostics	Coefficients, permutation importance, subgroup error and residual analysis.

18. Key Results Appendix

Topic	Key quantitative result
PM2.5 distribution	Mean 35.05; median 29.0; maximum 170.4
Seasonal medians	Winter 48.82; Pre-Monsoon 31.68; Monsoon 18.06; Post-Monsoon 29.46
Seasonal effect	Kruskal-Wallis H 420.076; epsilon-squared 0.430
Hourly persistence	Lag 1 = 0.965; lag 24 = 0.877; lag 168 = 0.687
Daily persistence	Lag 1 = 0.919; lag 7 = 0.656; lag 30 = 0.557
Extreme threshold	Study-wide PM2.5 p90 = 66.1; p95 = 83.6
Top-10% episodes	303 total; 254 sustained; median 6 h; maximum 65 h
Winter episode rate	31.37 starts per 1,000 observed hours
Particulate AQL	397 days >100; 145 days >150
Forecast baseline	Persistence test MAE 12.085; RMSE 16.284; R-squared 0.738

Final Ridge	Test MAE 11.314; RMSE 15.028; R-squared 0.777
Ridge skill	6.4% lower test MAE and 7.7% lower test RMSE than persistence
Ridge win rate	About 55.4% of individual test forecasts
Primary forecasting insight	Persistence wins under very stable conditions; Ridge gains value as 24-hour pollution change grows

19. References and Method Notes

U.S. Environmental Protection Agency. 2024 revision of PM2.5 AQI breakpoints and Air Quality Index reporting framework. The project uses the revised PM2.5 daily sub-index breakpoints for a consistent retrospective particulate-AQI classification.

Statistical implementations use SciPy and statsmodels, including ADF/KPSS/Ljung-Box diagnostics, nonparametric tests and HAC/Newey-West covariance estimates. Forecasting implementations use scikit-learn pipelines, TimeSeriesSplit, Ridge regression, Random Forest, Histogram Gradient Boosting and permutation importance.

The interactive application is implemented with Streamlit and Plotly and reads processed datasets and saved phase-output tables from the repository.

Project dashboard: <https://air-quality-intelligence-kathmandu.streamlit.app/>

Colab: <https://colab.research.google.com/drive/1ggNzvyVNOIOV7N-dsJRenPkeBff1CPgy#scrollTo=d76586c8>